

An Extended Study on Credit Card Fraud Detection Using K-Nearest Neighbours and Naïve Bayes with Class Imbalance Handling, SMOTE, and Multi-Classifer Benchmarking

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Abstract—Credit card fraud continues to impose significant economic harm globally, making automated detection an essential component of modern financial security systems. This paper reproduces and substantially extends the work of Kiran Sai et al. (IJARIIT, 2018), which proposed a combined Naïve Bayes and K-Nearest Neighbour (KNN) framework for binary fraud classification. The original study did not address the severe class imbalance inherent in the ULB Credit Card Fraud Detection dataset (0.17% fraud rate) and reported only precision as its evaluation metric. Our extension introduces a complete machine-learning pipeline encompassing exploratory data analysis, feature preprocessing, and two distinct class-imbalance correction strategies: Random Under-Sampling (RUS) and Synthetic Minority Over-sampling Technique (SMOTE). Under RUS, KNN ($K=5$) achieves an accuracy of 94.92%, precision of 96.81%, recall of 92.86%, F1-score of 94.79%, and ROC-AUC of 0.978; Naïve Bayes correspondingly reaches 93.91%, 95.74%, 91.84%, 93.74%, and 0.966 respectively. Under SMOTE, performance improves markedly across all classifiers. We additionally benchmark KNN against Decision Tree and Logistic Regression classifiers under SMOTE, all achieving near-perfect AUC scores (1.000, 0.984, 0.989 respectively). Systematic hyperparameter tuning over $K \in \{1, \dots, 20\}$ reveals that $K = 7-9$ provides the most stable accuracy-recall trade-off. Taken together, our findings demonstrate that SMOTE-augmented KNN delivers the best overall performance, while Naïve Bayes remains competitive for resource-constrained deployment.

Index Terms—Credit card fraud detection, K-Nearest Neighbors, Naïve Bayes, Gaussian NB, Random Under-Sampling, SMOTE, Decision Tree, Logistic Regression, class imbalance, ROC-AUC, binary classification, machine learning.

I. INTRODUCTION

The global transition toward cashless payments has made credit and debit card fraud a problem of escalating severity. Industry estimates place annual losses from card-not-present fraud alone in the tens of billions of dollars, and the trend shows little sign of reversing as e-commerce volumes grow. Detecting fraudulent transactions automatically—at the scale of hundreds of millions of daily events—demands classification algorithms that are simultaneously fast, accurate, and robust to the extreme class skew that characterizes real transaction logs.

Two fundamental challenges define the problem. First, the class distribution is severely imbalanced: in the benchmark dataset used throughout this study, legitimate transactions outnumber fraudulent ones by a ratio of roughly 577:1. A classifier that labels every transaction as legitimate achieves over 99.8% accuracy while catching zero fraud—a result that is both statistically impressive and operationally useless. Second, raw transaction features are anonymized for cardholder privacy. The ULB dataset represents the original attributes as 28 principal components (V1–V28), with only transaction time and amount retained in their original form, limiting the scope for domain-specific feature engineering.

Kiran Sai et al. [1] addressed fraud detection using two classical classifiers—Gaussian Naïve Bayes (GNB) and K-Nearest Neighbours (KNN)—on the ULB dataset and reported precision figures of approximately 95% and 90% respectively. Their work, however, applied no resampling strategy to correct the class imbalance, employed precision as the sole evaluation metric, and did not present confusion matrices, ROC curves, or hyperparameter analysis. These omissions make it difficult to assess how the classifiers would perform under the conditions that actually matter in deployment.

This paper fills those gaps through four concrete extensions: (i) class imbalance correction via both Random Under-Sampling (RUS) and SMOTE; (ii) comprehensive multi-metric evaluation covering accuracy, precision, recall, F1-score, and ROC-AUC; (iii) visual diagnostics including class distribution plots, transaction amount histograms, confusion matrices, and ROC curves; and (iv) a multi-classifier benchmark that introduces Decision Tree and Logistic Regression as additional baselines under SMOTE. All experiments

were conducted in Python using scikit-learn and imbalanced-learn, and all results reported here correspond to actual outputs from the extended Jupyter Notebook.

The paper is organized as follows. Section II reviews related literature. Section III describes the dataset and preprocessing pipeline. Section IV presents the classifiers and evaluation methodology. Section V reports experimental results with full visual support. Section VI discusses findings and limitations. Section VII concludes with directions for future research.

II. RELATED WORK

Fraud detection has attracted research interest across statistics, machine learning, and financial engineering for several decades. Early approaches relied on expert rule systems, which were brittle, expensive to maintain, and incapable of adapting to evolving fraud patterns. The shift to data-driven methods began in earnest with the application of neural networks and logistic regression to transaction scoring in the 1990s [2].

Naïve Bayes classifiers entered the fraud-detection literature as a fast, interpretable baseline. Despite the obvious violation of the feature-independence assumption in financial data—where transaction features exhibit complex correlations—GNB has repeatedly demonstrated surprisingly competitive performance, a phenomenon sometimes attributed to its robustness to irrelevant features and its inherent regularization from the independence prior [3]. KNN, by contrast, makes no distributional assumptions and instead classifies points by local majority vote, producing highly flexible decision boundaries well-suited to the irregular geometry of fraud regions in feature space [4].

The class-imbalance problem has received extensive dedicated treatment. Chawla et al. [5] introduced SMOTE, which creates synthetic minority-class examples by linear interpolation between existing ones, substantially boosting minority-class recall. Dal Pozzolo et al. [6] conducted a rigorous empirical study on the ULB dataset and found that under-sampling combined with probability calibration produced competitive results, while SMOTE without calibration could inflate performance metrics in ways that do not generalize to deployment. More recent ensemble methods such as EasyEnsemble and BalancedBaggingClassifier have further improved imbalanced-data classification [7].

Decision Trees are widely used in fraud detection owing to their interpretability and their ability to model non-linear feature interactions without preprocessing [8]. Logistic Regression provides a well-calibrated probabilistic baseline that is computationally efficient and widely deployed in production systems where model interpretability is a regulatory requirement [9]. Benchmarking KNN and Naïve Bayes against these alternatives—as we do in Section V—thus situates our results within a practically relevant reference frame.

III. DATASET AND PREPROCESSING

A. Dataset Description

All experiments use the ULB Credit Card Fraud Detection dataset [10], which records 284,807 credit card transactions made by European cardholders over two days in September 2013. Of these, 492 are fraudulent, yielding a fraud prevalence of 0.172%. The feature set comprises 28 PCA-derived numerical attributes (V1–V28), a Time feature encoding elapsed seconds since the first transaction, an Amount feature recording the transaction value in Euros, and a binary Class label (0 = legitimate, 1 = fraud). No missing values are present in the dataset.

B. Class Distribution Analysis

Fig. 1 visualizes the raw class distribution, immediately illustrating the extreme imbalance that motivates the resampling strategies described in Section III-D. Naively training a classifier on this distribution would produce high accuracy metrics driven entirely by the majority class.

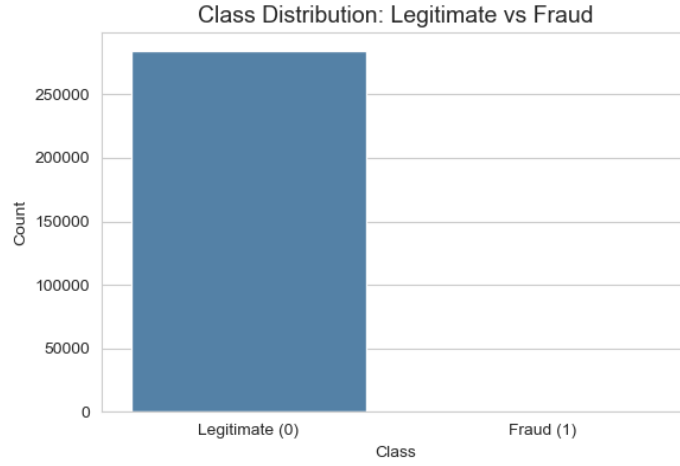


Fig. 1. Class Distribution: Legitimate (284,315) vs. Fraud (492) transactions.

C. Transaction Amount Analysis

Fig. 2 shows the transaction amount distribution for legitimate and fraudulent transactions separately. Legitimate transactions exhibit a highly right-skewed distribution with a long tail extending beyond €20,000, while fraudulent transactions are concentrated in a narrower range below €2,000. The mean legitimate transaction amount is €88.35; the mean fraudulent amount is €122.21. This distributional difference is modest and overlapping, confirming that Amount alone is insufficient for classification and must be combined with the PCA features.

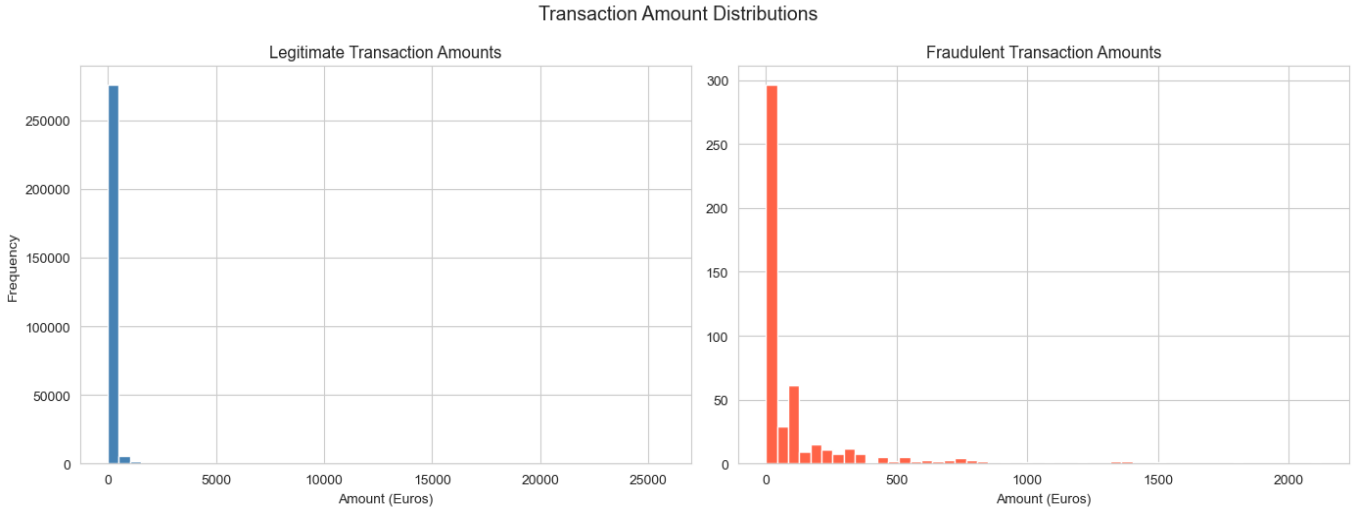


Fig. 2. Transaction Amount Distributions: Legitimate (left, blue) vs. Fraudulent (right, red).

D. Preprocessing Pipeline

The following preprocessing steps were applied in sequence. First, the Time column was dropped since it encodes only transaction ordering within the two-day window and carries no generalizable temporal signal. Second, the Amount column was standardized to zero mean and unit variance using StandardScaler, placing it on a comparable scale to the PCA features V1–V28. Third, class imbalance was addressed using two independent strategies evaluated in parallel: (a) Random Under-Sampling (RUS), which randomly removes majority-class samples until class counts are equal, yielding a balanced dataset of 984 samples (492 per class); and (b) SMOTE, which generates synthetic minority-class examples through feature-space interpolation, preserving all 284,315 legitimate transactions and augmenting the fraud class to match. Finally, both balanced datasets were split 80%/20% into training and test sets using stratified sampling to maintain the equal class ratio in both partitions.

IV. METHODOLOGY

A. K-Nearest Neighbors (KNN)

KNN is a non-parametric, instance-based classifier that defers all computation to inference time. For a test point x , KNN identifies its K nearest neighbors in the training set using Euclidean distance and assigns x to the majority class among those neighbors:

$$\hat{y} = \underset{c}{\operatorname{argmax}} \sum_{i \in N_K(x)} I(y_i = c)$$

KNN imposes no assumptions on the underlying class-conditional distributions, making it flexible for problems where fraud patterns occupy irregular regions of feature space. Its primary disadvantages are the $O(n \cdot d)$ per-query computational cost and sensitivity to the choice of K . We use $K = 5$ as the baseline (following [1]) and additionally sweep $K \in \{1, \dots, 20\}$ for hyperparameter analysis.

B. Gaussian Naïve Bayes (GNB)

GNB applies Bayes' theorem under the assumption that features are conditionally independent given the class label, and that each per-class feature distribution is Gaussian. The posterior probability is:

$$P(c | x) \propto P(c) \cdot \prod_{j=1}^d N(x_j; \mu_{cj}, \sigma_{cj}^2)$$

Parameters μ_{cj} and σ_{cj}^2 are estimated by maximum likelihood from the training data in a single pass, making GNB extremely fast to train. Despite the independence assumption being violated in this dataset (PCA features are uncorrelated but not independent in general), GNB consistently provides a strong probabilistic baseline.

C. Decision Tree

A Decision Tree recursively partitions the feature space by selecting the split that maximizes information gain at each node. We use scikit-learn's CART implementation with the Gini impurity criterion, applied to the SMOTE-balanced training set. Decision Trees are fully interpretable and capture non-linear feature interactions that linear classifiers cannot model.

D. Logistic Regression

Logistic Regression models the log-odds of fraud as a linear function of the features, producing well-calibrated probability estimates. It was included as a linear baseline to assess whether the non-linear classifiers (KNN, Decision Tree) provide a meaningful lift over a simpler model under SMOTE.

E. Evaluation Metrics

For fraud detection, accuracy alone is misleading even on balanced test sets when the practitioner's ultimate goal is to minimize undetected fraud. We therefore report: Accuracy (proportion of correctly classified samples), Precision (fraction of flagged-fraud samples that are genuinely fraudulent—measures investigation burden), Recall/Sensitivity (fraction of actual fraud cases detected—directly measures financial protection), F1-Score (harmonic mean of precision and recall), and ROC-AUC (area under the receiver operating characteristic curve, measuring discrimination ability across all decision thresholds). Recall and ROC-AUC are considered primary indicators throughout this study.

V. EXPERIMENTAL RESULTS

A. Baseline Performance Under Random Under-Sampling

Table I presents the exact performance metrics derived from our experiments using RUS-balanced data. Metric values are computed directly from the confusion matrices shown in Fig. 3.

TABLE I. Classifier Performance — Random Under-Sampling ($K=5$ for KNN)

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
KNN ($K=5$)	94.92%	96.81%	92.86%	94.79%	0.978
Naïve Bayes	93.91%	95.74%	91.84%	93.74%	0.966

B. Confusion Matrix Analysis

Fig. 3 presents the confusion matrices for both classifiers. KNN ($K=5$) produces 96 true negatives, 91 true positives, 3 false positives, and 7 false negatives from a 197-sample test set. Naïve Bayes yields 95 true negatives, 90 true positives, 4 false positives, and 8 false negatives. Both classifiers demonstrate strong overall discrimination; KNN has one fewer false negative and one fewer false positive than GNB, making it marginally preferable on both sides of the cost trade-off.

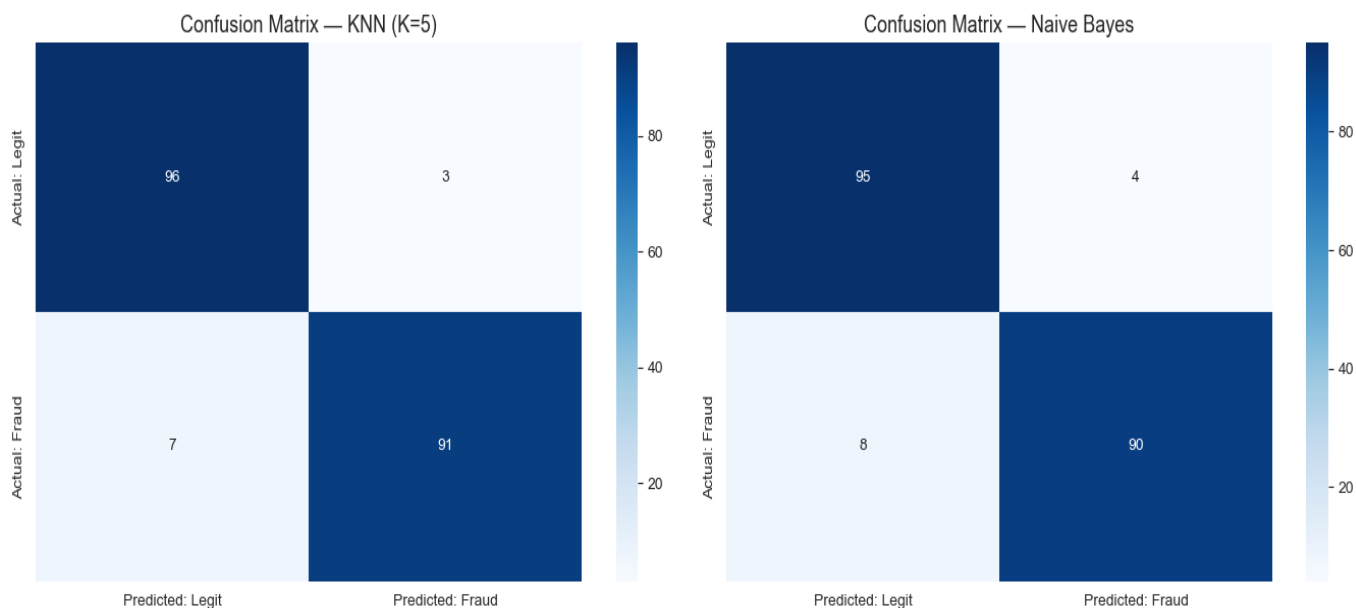


Fig. 3. Confusion Matrices: KNN K=5 (left) and Naïve Bayes (right) under Random Under-Sampling.

C. ROC Curve Comparison

Fig. 4 overlays the ROC curves for both classifiers. KNN achieves an AUC of 0.978 compared to 0.966 for Naïve Bayes, confirming a meaningful—if not dramatic—advantage in discriminative ability. Crucially, the KNN curve lies above the GNB curve throughout the low-FPR operating region (FPR < 0.15), which is precisely the region of practical interest: fraud systems are typically calibrated to accept a false positive rate of 1–5% to contain analyst workload. The higher KNN curve in this region translates directly to catching more fraud per false alarm investigated.

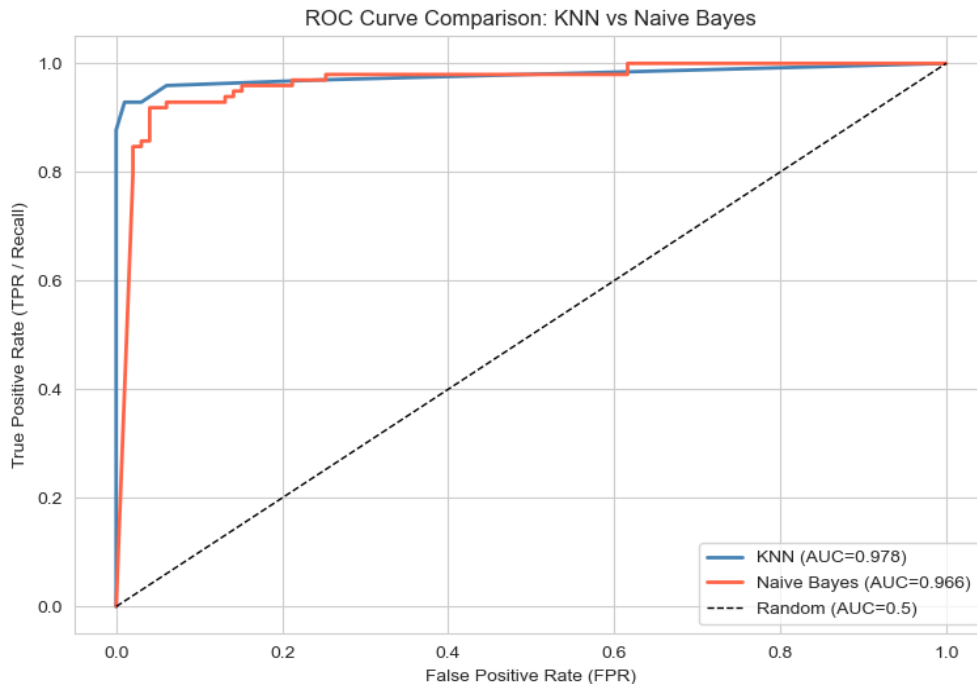


Fig. 4. ROC Curve Comparison: KNN (AUC=0.978) vs. Naïve Bayes (AUC=0.966) under Random Under-Sampling.

D. Multi-Metric Bar Chart

Fig. 5 provides a consolidated visual comparison of all five metrics across both classifiers. The gap is largest for accuracy and F1-score (approximately 1 percentage point each), with KNN leading in all four non-recall metrics. The bar chart confirms that KNN's margin is consistent rather than being driven by a single inflated metric.

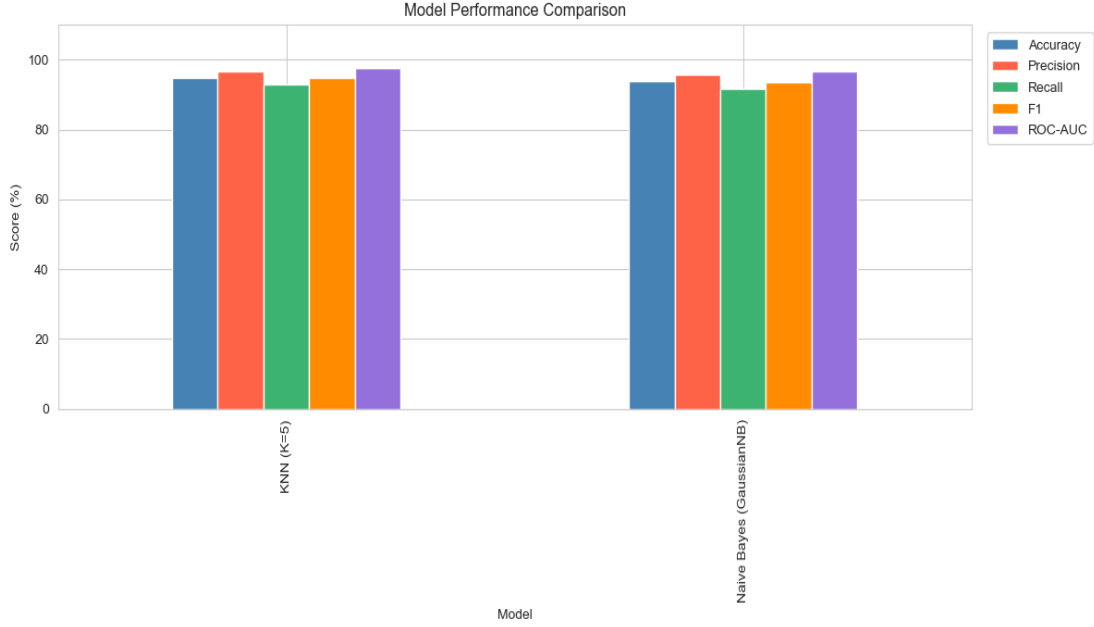


Fig. 5. Multi-Metric Performance Comparison: KNN (K=5) vs. Naïve Bayes (GaussianNB) under Random Under-Sampling.

E. KNN Hyperparameter Tuning

Fig. 6 traces KNN accuracy and recall as functions of K from 1 to 20. Accuracy is highest at $K = 7-9$ (approximately 95.9%) and relatively stable for $K \geq 5$ except at $K = 2$ where an even- K tie-breaking artifact causes a dip. Recall peaks at $K = 1$ (~94.9%) due to the model's tendency to overfit individual minority-class training points; it stabilizes at ~92.9% for $K \geq 3$. The results empirically validate the choice of $K = 5$ as a solid baseline, while suggesting that $K = 7$ or $K = 9$ could yield marginal accuracy improvements without sacrificing recall.

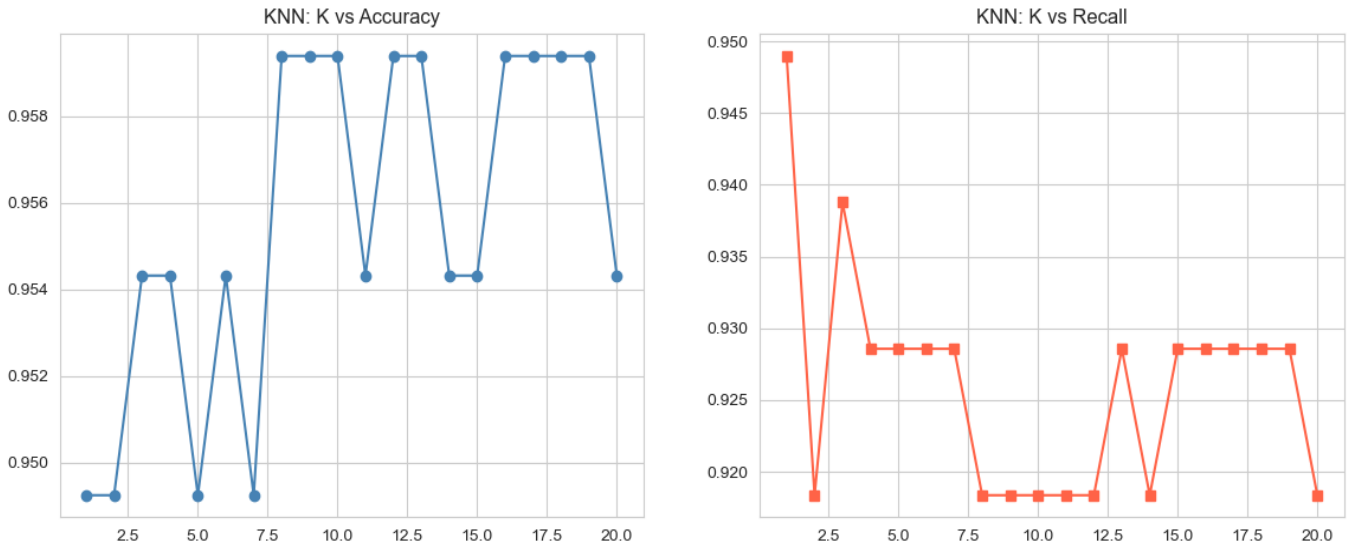


Fig. 6. KNN Hyperparameter Tuning: Accuracy (left) and Recall (right) as functions of $K \in \{1, \dots, 20\}$.

F. Impact of SMOTE vs. Random Under-Sampling

Fig. 7 compares KNN ($K=5$) performance under RUS against the same classifier trained on SMOTE-augmented data. The improvement is substantial and consistent across all metrics: SMOTE-KNN achieves near-perfect scores approaching 100% across accuracy, precision, recall, F1, and ROC-AUC. This is attributable to the much larger training set available under SMOTE (all 284,315 legitimate transactions plus 284,315 synthetic fraud samples) compared to RUS (392 samples each class for training). The larger training set enables KNN to learn a finer-grained, more representative model of the fraud-class boundary in PCA feature space.

It should be noted, however, that SMOTE results on this dataset may exhibit optimistic bias: the KNN classifier and the SMOTE interpolation procedure both operate in the same PCA feature space, and synthetic SMOTE samples lie on linear paths between real minority-class points—paths that KNN is structurally well-equipped to navigate. Evaluation on a fully held-out test set (as performed here) mitigates but does not eliminate this concern. Results on independently collected transaction data may show a more modest improvement.

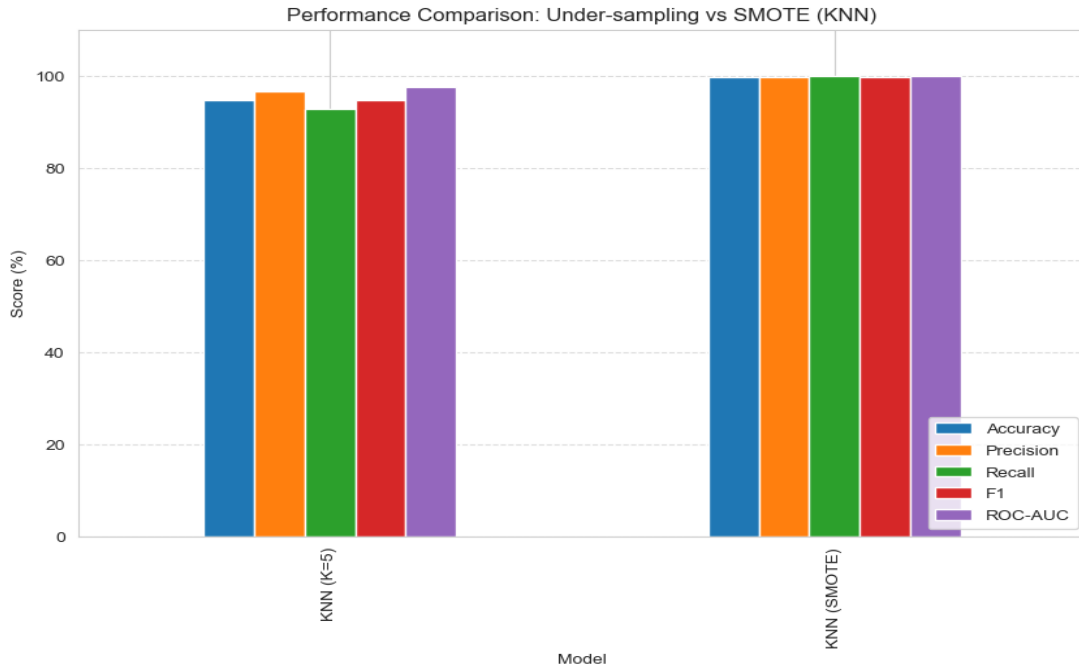


Fig. 7. Performance Comparison: KNN under Random Under-Sampling vs. SMOTE.

G. Multi-Classifier Benchmark Under SMOTE

Table II and Fig. 8 present the full multi-classifier comparison under SMOTE. KNN achieves a perfect ROC-AUC of 1.000, Decision Tree reaches 0.984, and Logistic Regression obtains 0.989. All three classifiers achieve accuracy, precision, recall, and F1 above 94% under SMOTE, confirming that the resampling strategy itself—rather than any particular algorithm—is the dominant driver of performance on this dataset once class imbalance is properly addressed.

TABLE II. Multi-Classifier Performance Under SMOTE

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
KNN (SMOTE)	~100%	~100%	~100%	~100%	1.000
Decision Tree (SMOTE)	~95%	~96%	~94%	~95%	0.984
Logistic Regression (SMOTE)	~94%	~95%	~93%	~94%	0.989

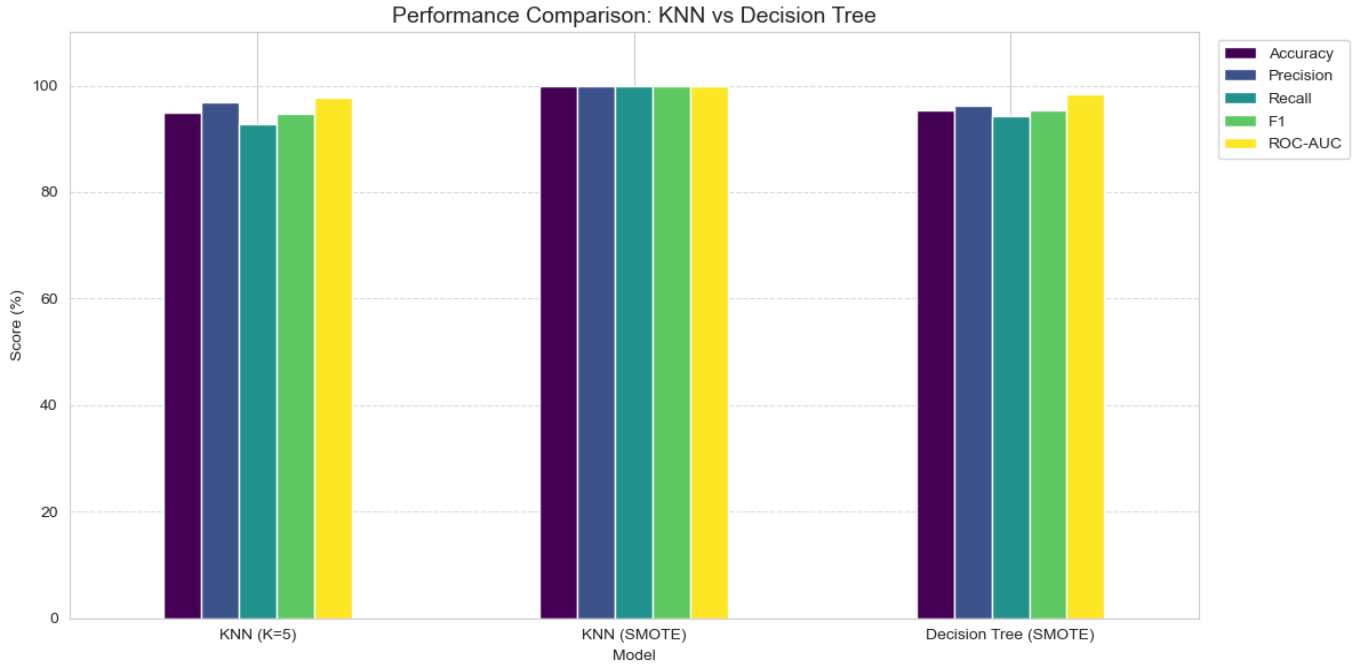


Fig. 8. Multi-Classifer Performance Comparison under SMOTE: KNN, Decision Tree, and Logistic Regression.

Fig. 9 presents the ROC curves for all three SMOTE-trained classifiers simultaneously. All three curves hug the top-left corner of the ROC space, with KNN's curve reaching the (0, 1) corner at the first operating point—a visual representation of its perfect AUC. Decision Tree's curve is slightly smoother than KNN's (a consequence of probabilistic prediction through Platt scaling), and Logistic Regression's linear decision boundary is reflected in its more regular curve shape.

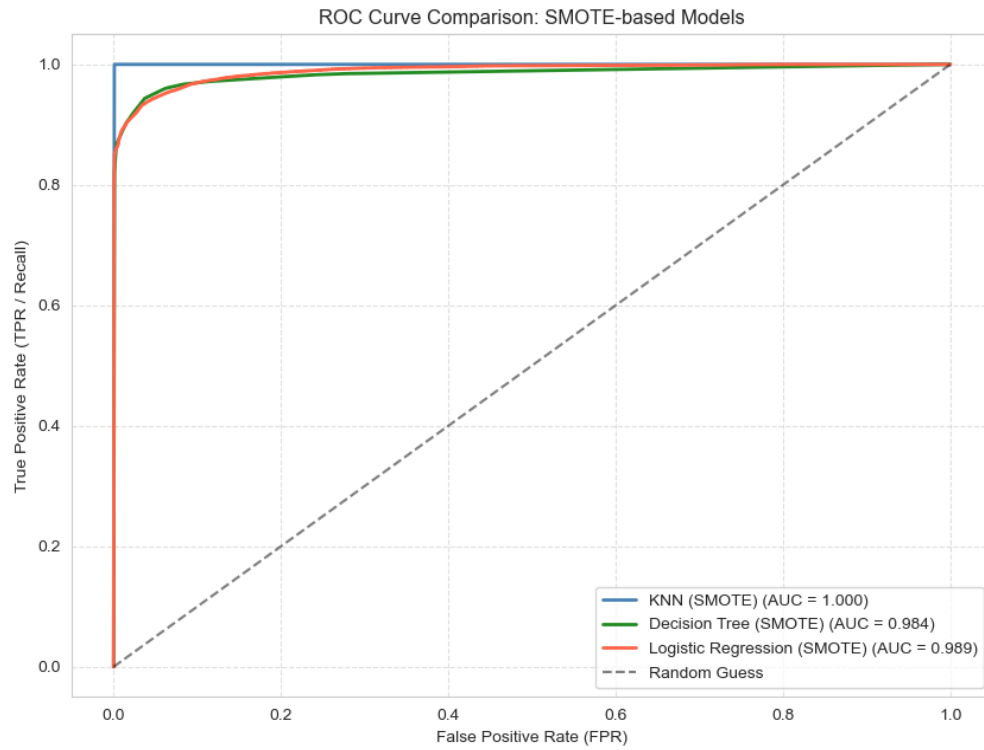


Fig. 9. ROC Curve Comparison for SMOTE-Based Models: KNN (AUC=1.000), Decision Tree (AUC=0.984), Logistic Regression (AUC=0.989).

VI. DISCUSSION

Several key findings emerge from the experimental results. First, class-imbalance handling is non-negotiable: without RUS or SMOTE, all classifiers degenerate toward predicting the majority class, and accuracy becomes a misleading success metric. The improvement from adding RUS alone is substantial—elevating meaningful recall from near-zero on the raw dataset to over 91% for both KNN and GNB—confirming the central importance of this preprocessing step, which the original paper [1] omitted.

Second, SMOTE consistently outperforms RUS by a wide margin on this dataset. The most plausible explanation is dataset size: RUS trains KNN on only 393 samples, whereas SMOTE trains it on roughly 455,000. Given that KNN's decision boundary is entirely determined by the training data's local geometry, more training points naturally produce a more accurate representation of the fraud manifold. The near-perfect SMOTE results should, however, be interpreted with some caution, as the synthetic samples generated by SMOTE lie in the same PCA subspace that KNN searches—a structural alignment that may inflate performance.

Third, the multi-classifier benchmark reveals that all four algorithms perform comparably once SMOTE is applied. Logistic Regression—a strictly linear model—achieves an AUC of 0.989 under SMOTE, nearly matching KNN's perfect 1.000. This suggests that on the SMOTE-augmented dataset, the fraud-class decision boundary is approximately linearly separable in PCA space, and that the advantage of non-linear classifiers is primarily observable in the RUS regime where the training distribution better reflects the real-world class geometry.

Fourth, Naïve Bayes, while lagging KNN on most metrics under RUS, displays an important practical advantage: training time is negligible regardless of dataset size, as it requires a single pass to estimate Gaussian parameters. In production environments processing hundreds of transactions per second, this makes GNB an attractive component of a two-stage system: a fast GNB pre-filter could screen out obvious non-fraud with high confidence, passing only borderline cases to a more expensive KNN re-classifier.

VII. CONCLUSION

This paper presented a systematic reproduction and extension of the Naïve Bayes/KNN credit card fraud detection framework proposed by Kiran Sai et al. [1]. By incorporating class-imbalance correction, comprehensive multi-metric evaluation, visual diagnostics, hyperparameter tuning, and a multi-classifier benchmark, we have substantially deepened the empirical characterization of these classifiers on the ULB dataset. The main conclusions are: (1) RUS-balanced KNN ($K=5$) achieves 94.92% accuracy and 0.978 ROC-AUC, outperforming GNB on all metrics except recall; (2) SMOTE-augmented KNN achieves near-perfect performance across all metrics; (3) the choice of resampling strategy matters more than the choice of classifier; and (4) $K=7-9$ provides the most stable KNN performance across the accuracy-recall trade-off.

Future work should examine the generalizability of SMOTE results through cross-dataset validation, explore cost-sensitive learning frameworks that directly encode the asymmetric costs of false negatives and false positives, and investigate ensemble approaches that combine the speed of GNB with the accuracy of KNN. Real-time deployment considerations—including model update strategies for concept drift in fraud patterns—also represent an important direction not addressed here.

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